

Dual-Display Timing Desk

Keep one stopwatch story coherent across two independent presenters

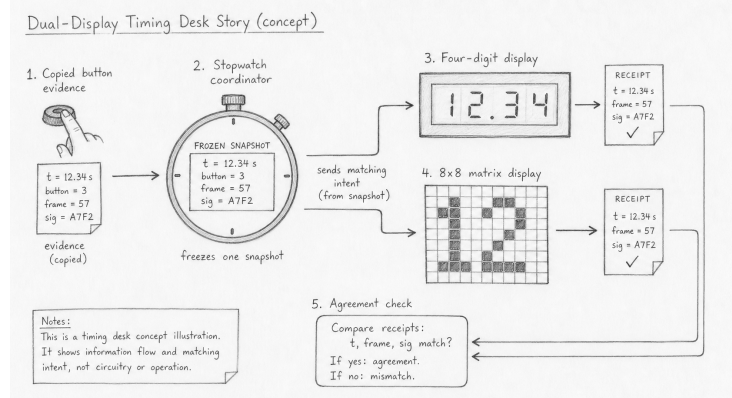
E0 COPIED POLICY REPLAY — ZERO DISPLAY HARDWARE

Lesson	060 — project	Time	180–240 minutes
Prerequisites	Lessons 058–059	Board	compile-only Mega replay
Capacity	one stopwatch and one pending presentation	Observation	named memory result cells

Responsibility. `DualDisplayTimingDesk` owns deterministic stopwatch state, copied control admission, one immutable presentation snapshot, two generation-bound display intents, self-test progress, and disagreement provenance. It owns no button, pin, timer, interrupt, bus, display, module, supply, clock, retry loop, or powered circuit.

1 Build one story, not two clocks

The timing desk presents the same stopwatch snapshot in two different ways. Four digits show minutes, seconds, and tenths. An eight-by-eight matrix shows a perimeter dial and a central status glyph. Those outputs are generated independently, so Lesson 060 must preserve the shared meaning before either child policy begins its own service sequence.



Alternative text: a pencil-drawn stopwatch snapshot card forks into a four-digit time card and an eight-by-eight dial card; both cards carry the same nonzero presentation generation while two independent service paths continue below.

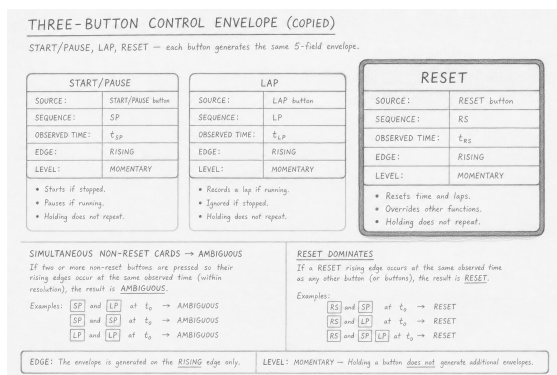
Atomic presentation means that both children receive values derived from one immutable logical snapshot. It does not mean simultaneous photons, a rollback-capable MAX7219 transport, or proof that either display changed.

E0 proves	E0 does not prove
one copied stopwatch snapshot feeds both intents	both displays changed at the same instant
both previews pass before either pure commit	physical writes can roll back
accepted logical frame receipts match expected values	LEDs were visible or optically agreed
bounded supplied-time behavior	an interrupt or hardware timer met deadlines

2 Admit copied controls before changing time

The project consumes copied start/pause, lap, and reset evidence. Each envelope must carry the expected source identity, configuration revision, sequence, observation time, and consistent edge and level facts. Structural validation happens before reset precedence, so a reset bit cannot smuggle malformed companion evidence into the state machine.

Reset dominates every valid simultaneous control. Without reset, simultaneous start/pause and lap presses are ambiguous and do not mutate the stopwatch. Held levels do not repeat; only admitted edges act.



Alternative text: three pencil button cards enter a validation gate carrying identity, sequence, time, edge, and level fields; after the gate, reset takes the upper dominant path while a simultaneous non-reset pair enters an ambiguity box without changing stopwatch state.

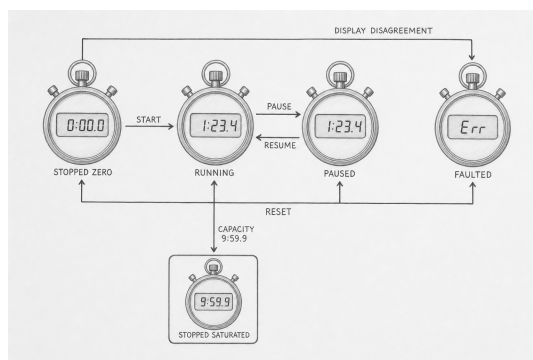
Valid edge set	Result	Reason
reset plus anything	stopped at canonical zero	reset dominance
start/pause only	start, pause, or resume	phase-dependent toggle
lap only while running	copy current elapsed for a timed hold	stopwatch continues internally
start/pause plus lap	no stopwatch mutation	ambiguous non-reset intent
no edge; held level	no repeated action	edges act once

Before replay, predict whether a structurally invalid envelope containing a reset edge can clear a fault: _____.
 Then explain which validation rule supports your answer: _____.

3 Derive elapsed time only from supplied time

The stopwatch phases are stopped-zero, running, paused, and faulted. Running elapsed time is derived from supplied **TimePoint** values. Pause materializes elapsed time; resume continues from that materialized value. Lap copies the current elapsed value without stopping the internal clock.

Capacity is 9:59.9. Reaching capacity saturates visibly and stops rather than wrapping. Backward time and an exact unsigned half-range difference reject atomically.



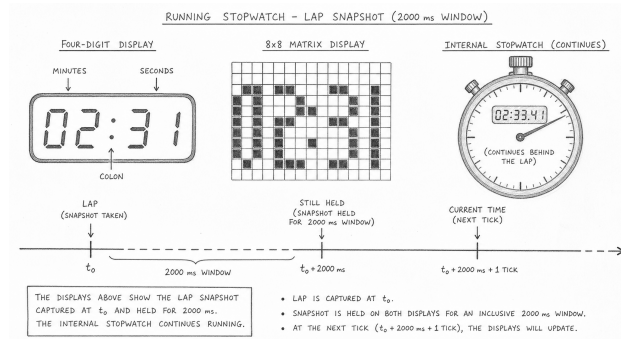
Alternative text: a hand-drawn state map connects stopped zero, running, and paused with start-pause edges; a lap tracing sheet overlays running without stopping it, reset returns every valid state to zero, and the capacity boundary leads to a saturated stopped card.

Boundary	Prediction	Evidence to inspect
start at zero	running with zero elapsed	phase and anchor time
pause at 12,345 ms	paused at copied elapsed	materialized elapsed
resume, then lap	running continues behind frozen presentation	current and lap snapshots
reach 599,900 ms	show 9:59.9, then stop	saturation and phase
time regression	no accepted state changes	control status and prior snapshot

4 Map the ordinary digit frame

Elapsed time renders left to right as minute, seconds tens, seconds units, and tenths. Decimal mask 0x0A lights the separator after the minute and the separator after the seconds units. Stopped and running show current elapsed time; paused shows materialized elapsed.

A lap press freezes both presentation frames at the lap snapshot for exactly 2,000 ms while a running stopwatch continues internally. The inclusive deadline still shows the lap. The next valid tick returns presentation to current time.

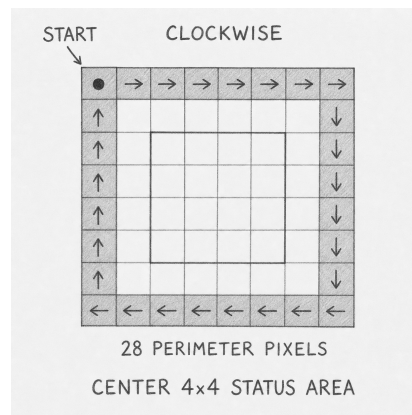


Alternative text: a pencil timeline shows a lap snapshot pinned above the two display cards through an inclusive two-second deadline while a separate stopwatch line keeps moving; the next tick releases the pin and freezes the newest current snapshot.

Elapsed	Four cells	Separator mask
0 ms	0:00.0	0x0A
1 ms	0:00.0	0x0A
61,200 ms	1:01.2	0x0A
599,900 ms or greater	9:59.9	0x0A

5 Draw the matrix dial from the same snapshot

The matrix dial uses 28 perimeter pixels clockwise from the top-left corner: across the top, down the right edge, back across the bottom, and up the left edge without repeating corners. For $ms = elapsed \bmod 60000$, zero milliseconds lights no perimeter pixel. Otherwise the count is $1 + \lfloor ms \times 28 / 60000 \rfloor$, capped at 28.



Alternative text: a pencil eight-by-eight grid numbers the 28 unique perimeter pixels clockwise from the top-left corner, with arrows at each corner and the central four-by-four area reserved for a status glyph.

The first millisecond lights the first pixel. Every bucket boundary is fixed by integer arithmetic. The final bucket lights all 28 pixels before minute rollover; rollover returns the perimeter to zero.

Time within minute	Lit perimeter count	Prediction to check
0 ms	0	completely empty perimeter
1 ms	1	top-left pixel only
first bucket boundaries	golden integer counts	no floating-point drift
last millisecond	28	complete perimeter
60,000 ms	0	minute rollover

The central rows and columns two through five carry one four-by-four status glyph. Its row nibbles are:

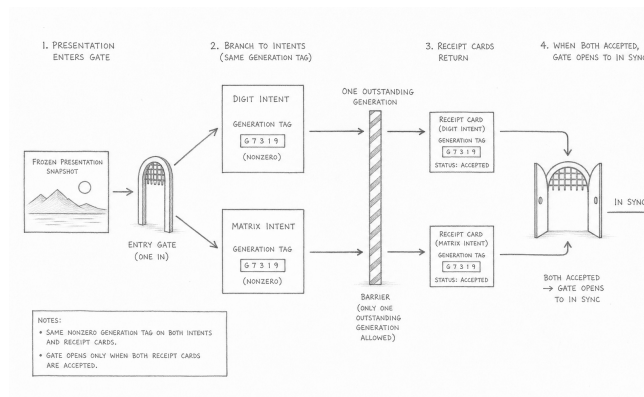
State	Four left-to-right row nibbles
stopped	F, 9, 9, F
running	6, 3, F, 2
paused	A, A, A, A
lap	8, 8, 8, F
fault	9, 6, 6, 9

Lesson 059 orientation is applied only after the complete logical perimeter and status frame exist.

6 Publish one generation and wait

Only one presentation generation may be outstanding. When none is pending, the coordinator freezes the newest post-control stopwatch snapshot, derives both frames, preflights both child policies, and commits both pure candidates. The nonzero generation and its grace interval begin after both commits and before the first child service.

While that generation is pending, supplied-time updates continue the internal stopwatch. Presentation reports **ResourceBusy**; neither frame intent is overwritten, and the caller must not replay an already admitted control edge.



Alternative text: one pencil generation card occupies a single-slot gate and feeds separate digit and matrix acceptance checkboxes; new stopwatch samples continue into a waiting latest-snapshot card but cannot replace the occupied presentation card.

```
result = desk.update(envelope);
control = result.controlStatus;
presentation = result.presentationDisposition;
digits = result.digitTransaction;
matrix = result.maxCommand;
```

`controlStatus` and `presentationDisposition` are deliberately separate. A presentation backlog must never make the caller repeat a control edge that was already admitted.

7 Correlate full receipts, not convenient hashes

Each copied endpoint receipt carries owner, lifecycle, configuration, requested and accepted generation, complete reported logical frame, reported frame digest, status, blank-request acceptance, and observation time. Generations are nonzero `uint32_t` values with modular half-range ordering and fault before wrapping to zero.

The digit and matrix sides use separate 32-bit FNV-1a digests. The domains are `ADK.DIGIT.FRAME.V1` and `ADK.MATRIX.FRAME.V1`, encoded as ASCII without a trailing NUL. The digit stream then alternates the one-byte numeric glyph and one-byte diagnostic glyph for digit indices zero through three, followed by the decimal mask, an overflow byte (0x00 or 0x01), the source snapshot sequence, frame generation, and presentation mode. The matrix stream instead follows its domain with rows zero through seven, then the same source sequence, frame generation, and mode. Both 32-bit integers enter least-significant byte first; the mode is one byte. FNV-1a begins at 0x811c9dc5, folds each byte separately, and multiplies by 0x01000193 after each XOR.

```
digit glyphs: {0, 1, 2, 3}
diagnostic glyphs: {1, 2, 3, 4}
decimal / overflow: 0x0A / true
```

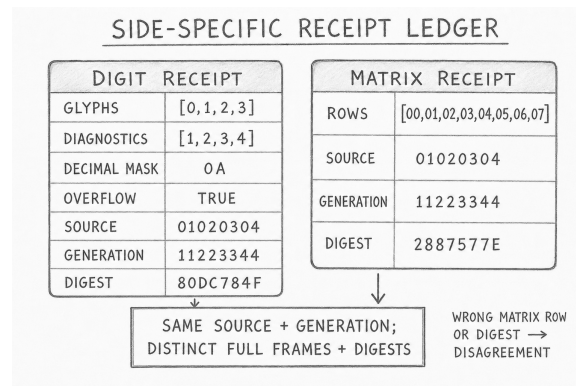
```

source / generation / mode:
  0x01020304 / 0x11223344 / 0x80
digit digest: 0x80dc784f

matrix rows: {0, 1, 2, 3, 4, 5, 6, 7}
matrix digest: 0x2887577e

```

The matrix witness uses the same source sequence, generation, and mode. These literal results freeze both field order and little-endian integer encoding.



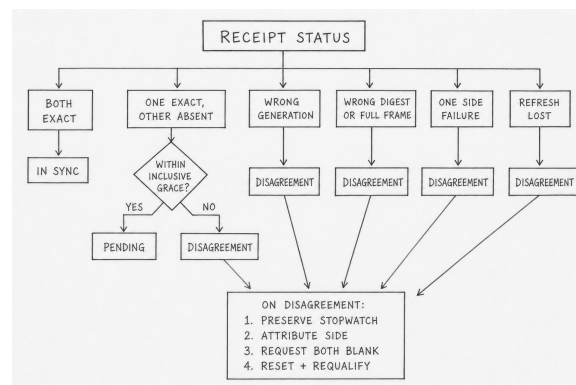
Alternative text: separate pencil ledgers retain the complete digit glyph, diagnostic, decimal-mask, and overflow frame and the complete matrix row frame. Both carry source sequence 01020304 and generation 11223344, while their side-specific digests are 80DC784F and 2887577E. A wrong matrix row or digest leads to disagreement.

Digest rule. A digest is compact redundant evidence, not structural identity. Acceptance requires the current generation, the side's expected digest, and byte-for-byte agreement with the retained full frame.

8 Bound grace and name disagreement

presentationGrace defaults to 100 ms and must be between 1 and 1,000 ms, well below modular half range. The generation remains pending through the inclusive deadline and expires on the next valid tick.

InSync requires both sides to accept their own current generation, digest, and complete frame. One accepted side and one pending side remains pending during grace. Wrong digest, stale or crossed generation, one-sided failure, digit refresh loss, or grace expiry enters presentation disagreement with side-specific provenance. Stopwatch history is preserved, but both displays receive blank requests.

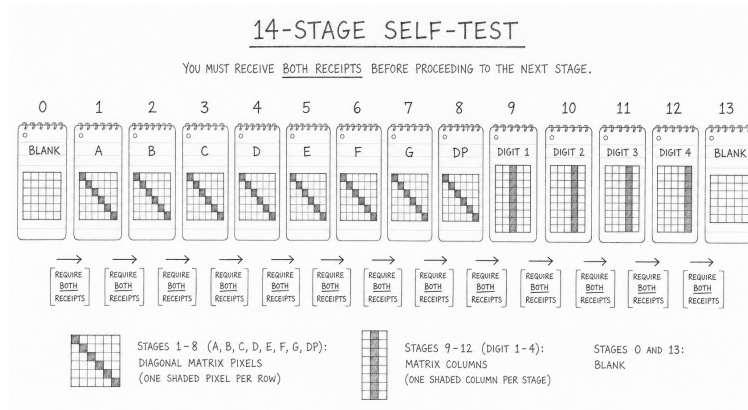


Alternative text: a pencil receipt-status decision tree keeps one exact side and one absent side pending through the inclusive grace deadline, reaches in-sync only through two exact matches, and routes late, wrong-generation, wrong-frame, wrong-digest, failed, or refresh-lost evidence to disagreement. Only disagreement requests both displays blank and begins reset and requalification.

Evidence	Disposition	Next observation
both exact current receipts	in sync	matching side generations and frames
one exact, one absent within grace	pending	inclusive deadline and missing side
wrong digest but matching full frame	disagreement	digest domain and byte order
right digest but changed full frame	disagreement	retained structural frame
digit refresh lost	disagreement	digit fault provenance and blank intent
next tick after grace	disagreement	anchored publication time

9 Earn Ready through the common self-test

Initialization does not jump directly to ordinary presentation. The deterministic self-test has fourteen common stages: blank; eight indexed segment-and-pixel stages; four digit-select-and-column stages; and final blank. A stage advances only after both complete correlated receipts match its generation and expected side-specific frame.



Alternative text: a pencil strip of fourteen numbered cards shows initial blank, eight cards pairing segment a through decimal point with matching diagonal matrix pixels, four cards pairing one digit showing eight with one matrix column, and final blank before a Ready gate.

During the eight indexed stages, the digit policy lights segment *a* through *dp* on digit *index* mod 4, while the matrix policy lights pixel (*index*, *index*). During the four select stages, one digit shows 8 while the matrix lights its matching column. **selfTestTimeout** is 100 ms per stage: inclusive at the deadline, faulting on the next valid tick. Ready zero is the first ordinary generation.

Predict which claim this E0 self-test supports: _____. Then name the additional E1c observation required before saying the two physical displays agree: _____.

10 Follow one bounded service opportunity

Every **update(envelope)** performs bounded work in a fixed order:

1. validate the complete copied envelope;
2. ingest child receipts;
3. service one due digit transaction;
4. service at most one MAX command;
5. evaluate grace and self-test;
6. apply valid control edges once;
7. advance the internal stopwatch; and
8. if no generation is outstanding, freeze, preflight, commit, and publish the next dual generation.

Structural rejection mutates nothing. The caller must still service the project often enough to satisfy Lesson 058's four-millisecond maximum digit gap. No loop catches up missed scans or submits several MAX commands at once.

11 Run the canonical replay

Open `examples/Lesson060DualDisplayTimingDesk/`
`Lesson060DualDisplayTimingDesk.ino`. Follow its narrative order: **acquire** copied fixtures, **configure** timing and orientation, **start** the self-test, then repeatedly **observe**, **decide**, and **present** at most one bounded service result.

Before running, record four predictions:

1. The number of common self-test stages is _____.
2. A valid lap edge is/is not replayed while presentation is busy: _____.
3. Grace faults at the inclusive deadline/on the next tick: _____.
4. The first ordinary Ready generation represents elapsed _____.

After running, inspect named volatile result cells. Serial output may narrate the trace, but it is never the sole evidence.

Observation cell	Expected result	Interpretation
self-test result	all fourteen stages correlated	Ready was earned logically
control result	admitted edge acts once	presentation busy is independent
digit result	one ordered refresh transaction at most	bounded scan service
matrix result	one register command at most	caller-paced transport
agreement result	two exact side receipts or named fault	no digest-only acceptance
shutdown result	both blank requests retained	logical safe intent only

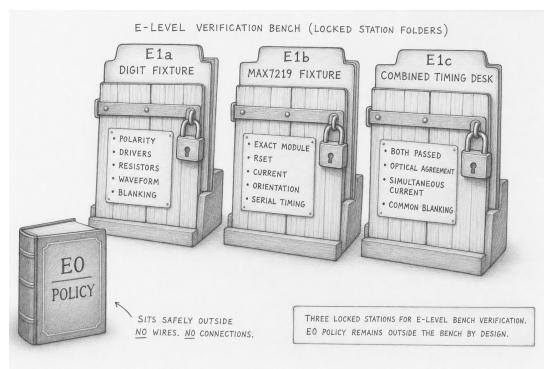
12 Diagnose the evidence boundary

Observed evidence	Likely policy meaning	Next check
control OK, presentation busy	edge admitted while prior generation remains	do not replay control; service children
self-test stage does not advance	one receipt absent or mismatched	compare generation and full expected frames
digit side faults first	refresh deadline or receipt evidence failed	inspect digit provenance and service cadence
matrix side faults first	row sequence or copied receipt failed	inspect first MAX operation failure
both digests match, agreement fails	generation or structural frame differs	compare every receipt field
blank requested after fault	disagreement policy acted	do not claim physical darkness

13 Keep the physical acceptance card open

This lesson proves copied controls, supplied-time stopwatch behavior, coherent dual frame generation, bounded child service, full-frame and digest correlation, grace, self-test, disagreement provenance, restart, shutdown, and compile-only Mega integration. It proves no electrical or optical behavior.

E1a must separately qualify the exact four-digit display, polarity, segment and digit drivers, one resistor per segment, peak and aggregate current, multiplex waveform, ghosting, and blank-on-release. E1b must separately qualify the exact MAX7219 IC or module, schematic, RSET, decoupling, matrix orientation, serial timing, current, startup, and shutdown. E1c may begin only after both sides pass independently; it then measures combined supply demand, timing interference, optical agreement, self-test visibility, fault blanking, and simultaneous-current behavior.



Alternative text: three pencil laboratory cards remain stamped OPEN: E1a qualifies the multiplexed digits, E1b qualifies the MAX7219 matrix, and only arrows from both accepted cards permit E1c combined timing-desk optical and current measurements.

Gate	Predict and observe	Acceptance record
E1a resource	exact digit resources acquire and roll back	_____
E1a safe state	selects and segments release safely	_____
E1b resource	exact bus/device resources acquire and roll back	_____
E1b safe state	qualified shutdown and power-removal state	_____
E1c agreement	both displays visibly encode one snapshot	_____
E1c fault	one-sided failure drives observed safe response	_____

14 Exit ticket

1. Why does one immutable presentation generation not promise simultaneous photons?
2. Why are control admission and presentation disposition separate?
3. What happens internally while a lap frame remains visible?
4. Why must a receipt match both its digest and complete retained frame?
5. When does the inclusive presentation grace interval expire?
6. Which evidence must pass before the combined E1c timing desk may be powered?

Completion evidence. You can replay every stopwatch phase, predict both frame mappings, trace the fourteen-stage self-test, distinguish control admission from presentation backlog, diagnose side-specific disagreement, and explain why copied logical acceptance is not optical or electrical proof. Every physical acceptance card remains explicitly open.

Canonical companions: [Lesson 060 sketch](#) [HTML lesson source](#) [implementation plan](#)